

▣ Differentiation of Binary cross entropy loss function.

Here

Binary Cross entropy loss-

$$L = -[y \log(a) + (1-y) \log(1-a)]$$

$$\therefore \frac{dL}{da} = -\frac{d}{da}[y \log(a) + (1-y) \log(1-a)]$$

$$\therefore \frac{d}{da}[y \log(a)] = y/a$$

$$\frac{d}{da}[(1-y) \log(1-a)] = -\frac{1-y}{1-a}$$

$$\therefore \frac{dL}{da} = -\left(y/a - \frac{1-y}{1-a}\right) = -y/a + \frac{1-y}{1-a}$$

$$= \frac{-y + ay + a - ay}{(1-a) \cdot a}$$

Here

$$= \frac{a-y}{a(1-a)}$$

$a = a(z)$ (Sigmoid Output)-

$$da/dz = a(1-a)$$

According to chain rule-

$$dL/dz = dL/da \times da/dz$$

$$= \frac{a-y}{a(1-a)} \times a(1-a)$$

$$= (a-y) \text{ Answer.}$$